

Analysis on Heterogeneous SSD Configuration with Quadruple-Level Cell (QLC) NAND Flash Memory

Yoshiki Takai¹, Mamoru Fukuchi¹, Reika Kinoshita¹, Chihiro Matsui² and Ken Takeuchi^{1,2}

¹Department of Electrical, Electronic and Communication Engineering, ²Research and Development Initiative
Chuo University, Tokyo, Japan, E-mail: takai@takeuchi-lab.org

Abstract—This paper investigates optimal heterogeneously-integrated SSD configuration [1] with various non-volatile memories including quadruple-level cell (QLC) NAND flash [2] considering SSD performance, energy consumption and SSD endurance lifetime. Based on the design methodology of heterogeneous SSD composed of storage class memory (SCM), multi-level cell (MLC) and triple-level cell (TLC) NAND flash [1], this paper newly adds QLC and single-level cell (SLC) NAND flash as memory components and re-optimizes the memory configurations for various workloads. Four types of SSD are compared; 1) SLC/QLC NAND flash, 2) SCM/QLC NAND flash, 3) SCM/TLC/QLC NAND flash and 4) SCM/TLC NAND flash. As a result, performance of SCM/QLC NAND flash is about one-fourth of that of SCM/TLC NAND flash for all workloads because QLC NAND flash has 5.7 times longer write latency than TLC NAND flash. For cold workloads or high total SSD cost assumption, SCM/TLC NAND flash is recommended. For hot workloads with low total SSD cost assumption, however, SLC/QLC NAND flash is optimal with emphasis on better SSD endurance lifetime. Finally, SCM/TLC/QLC NAND flash is the best when considering performance and energy consumption. In particular, for write-hot application, prxy_0, SCM/TLC/QLC NAND flash achieves 67% higher IOPS/cost [1] than SCM/TLC NAND flash.

Keywords—QLC NAND flash memory; storage class memory; heterogeneous SSD;

I. INTRODUCTION

To expand the capacity and reduce bit cost of NAND flash memory, number of bits per cell is increasing and quadruple-level cell (QLC: 4 bits/cell) NAND flash memory has been put into market [3, 4]. QLC NAND flash has the lowest cost and the longest latency as shown in Fig. 1. In addition, as the number of bits per cell increases, the threshold voltage (V_{TH}) margins between V_{TH} states become smaller. Due to the smaller V_{TH} margins, QLC NAND flash is sensitive to program disturb and data-retention errors. To suppress the interference among neighboring memory cells, write unit of QLC NAND flash is block which is 16.8 Mbytes, while other NAND flash can be written with a word-line unit as shown in Fig. 2 [5]. Because of the larger write unit, QLC NAND flash needs non-volatile (NV) cache memory which is faster and has smaller write unit.

Today, single-level cell (SLC: 1 bit/cell) NAND flash is used as NV cache memory for QLC NAND flash [3, 4]. Storage class memory (SCM) is also considered as NV cache memory for QLC NAND flash because SCM has much shorter latency and better endurance than SLC NAND flash. Furthermore, SCM does not need garbage collection (GC) like NAND flash memory as shown in Fig. 3. Thus, SCM and NAND flash heterogeneous SSD achieves much higher performance, better endurance and lower energy consumption than NAND flash only SSD [6]. As candidates for SCM, resistive RAM (ReRAM) [7], magnetic RAM (MRAM) [8] and phase change RAM (PRAM) [9, 10] are developed. In this paper, MRAM is assumed for SCM. Then, SLC/QLC NAND flash and SCM/QLC NAND flash heterogeneous SSD are evaluated to explore which of SLC or

SCM is suitable for NV cache of QLC NAND flash. As a reference or base-line of the comparison, small capacity SCM and cost-effective triple-level cell (TLC: 3 bits/cell) NAND flash are also combined to QLC NAND flash.

II. SYSTEM ARCHITECTURE OF HETEROGENEOUS SSD

In this paper, data are categorized into “hot”, “cold” and “frozen” defined as “frequently accessed”, “infrequently accessed” and “hardly overwritten” data, respectively. To manage these data efficiently, heterogeneous SSD uses two types of data management algorithms: non-volatile write-back (NV-WB) cache algorithm [11] for SCM and NAND flash heterogeneous SSD and round-robin frozen data collecting algorithm (RR-FDCA) [12] for two types of NAND flash heterogeneous SSD, top and bottom NAND, as shown in Fig. 4(b) and (c).

Fig. 4(a) shows NV-WB cache algorithm. This algorithm stores hot data in SCM and cold data in NAND flash. Furthermore, all write data from host are written to SCM. When SCM becomes almost full, the least recently used (LRU) data in SCM are evicted to NAND flash. When data are read from NAND flash, the data are copied to SCM. Fig. 4(b) shows RR-FDCA. RR-FDCA classifies hot or cold data and frozen data. Top NAND flash is faster and has better endurance than bottom NAND flash. All write data from host are written to top NAND flash. RR-FDCA evicts frozen data from top NAND flash to bottom NAND flash during GC of top NAND flash. For SCM/TLC/QLC NAND flash tri-hybrid [13], the algorithm combining NV-WB and RR-FDCA is applied (Fig. 4(c)).

III. EVALUATION SETTINGS

SSD performance, energy consumption and SSD endurance lifetime are evaluated in four configurations of heterogeneous SSD: 1) SLC/QLC NAND flash, 2) SCM/QLC NAND flash, 3) SCM/TLC/QLC NAND flash and 4) SCM/TLC NAND flash [14]. In this paper, the heterogeneous SSD uses QLC NAND flash as main storage. The capacity of QLC NAND flash is more than 50% of the total SSD capacity. The total SSD capacity is defined as the user data size + 25% over-provisioning. SCM capacity is limited to at most 9% of total SSD capacity to suppress the overall SSD cost. Characteristics of each memory are shown in Table 1 [2, 8, 15, 16, 17]. Though bit cost of SCM is 20 times higher than that of TLC NAND flash, SCM has 10^3 times shorter latency than SLC NAND flash. In addition, QLC NAND flash has 5.7 times longer write latency but only 25% lower bit cost than TLC NAND flash.

In this paper, a transaction-level-modeling based emulator [6] with one-week-long workloads of enterprise servers [18] are used to evaluate the SSD performance. The workloads are classified in four types according to write/read total size ratio and their data access frequencies. As shown in Table 2, four typical workloads are chosen from each type: prxy_0 (write-hot), hm_0 (write-cold), prxy_1 (read-hot) and src2_1 (read-cold). Note that of prxy_0 and hm_0 are mainly analyzed because write and read intensive workloads show similar characteristics.

IV. EVALUATION RESULTS AND DISCUSSION

A. Performance evaluation of four configurations of heterogeneous SSD

Fig. 5 shows Input/Output per second (IOPS) and IOPS/cost and Fig. 6 shows energy consumption of four configurations of heterogeneous SSD for prxy_0 and hm_0. IOPS results are normalized by those of SLC NAND flash 50% and TLC NAND flash 50% hybrid configuration. Energy consumption is approximately inversely proportional to IOPS. In the same SCM capacity, IOPS of SCM/QLC NAND flash is about one-fourth of that of SCM/TLC NAND flash for all workloads. This is because QLC NAND flash has 5.7 times longer write latency than TLC NAND flash.

For prxy_0, though TLC NAND flash has lower performance than SLC NAND flash, SCM/TLC/QLC NAND flash achieves higher IOPS than SLC/QLC NAND flash thanks to SCM NV cache that needs no GC. Fig. 7 shows simulated IOPS results for prxy_0 when total SSD cost can be about 1.5 times of TLC NAND flash only SSD. For performance comparison, SCM capacity is the same in each configuration. When comparing SLC/QLC NAND flash with SCM/QLC NAND flash, though IOPS of SLC/QLC NAND flash declines at the beginning due to GC of SLC NAND flash, the performance finally exceeds SCM/QLC NAND flash. Fig. 8 indicates the overhead time of eviction and GC during one-week. In SCM/QLC NAND flash, eviction to QLC NAND flash and GC in QLC NAND flash cause large overhead because 3% of SCM capacity is too small to store all hot data in SCM. However, in SLC/QLC NAND flash, GC overhead of QLC NAND flash is small because 30% capacity of SLC NAND flash can keep hot data and prevent GC of QLC NAND flash. In addition, SCM/TLC/QLC NAND flash achieves up to 67% higher IOPS/cost than SCM/TLC NAND flash when SCM capacity is less than 9%. This is because FDCA selectively stores hot data overflowed from SCM and frozen data to TLC and QLC NAND flash, respectively. As a result, GC overhead of TLC NAND flash is reduced to 17.7% as shown in Fig. 8. When SCM capacity is 9%, SCM/TLC NAND flash achieves the best performance. This is because most of hot data are stored in SCM and NAND flash is filled with cold or frozen data.

For hm_0, SCM 1% and TLC NAND flash 99% hybrid configuration achieves the highest IOPS/cost. The reason is that the number of cache hit in NV cache is small and NV cache just works as the FIFO buffer. Further, there is no merit to divide NAND flash into two parts such as TLC and QLC NAND flash because almost no hot data exist in NAND flash. Except for SCM/TLC NAND flash, SCM 1%, TLC NAND flash 49% and QLC NAND flash 50% tri-hybrid configuration achieves the highest IOPS/cost. This is because performance of NV cache has little effect on SSD performance.

B. Memory endurance lifetime evaluation of four configurations of heterogeneous SSD

Fig. 9 shows write/erase (W/E) cycles and Fig. 10 shows memory endurance lifetime of each memory for prxy_0 and hm_0. In this paper, memory endurance lifetime is defined as acceptable W/E cycle divided by W/E cycle in Fig. 9. SCM has 10^7 - 10^9 times higher acceptable W/E cycle and thus achieves much longer memory endurance lifetime than NAND flash. For prxy_0, W/E cycle of TLC NAND flash in Example 1 is 93% less than that of SLC NAND flash in Example 2. This is because write accesses concentrate on SCM. However, in Example 1 TLC NAND flash in Example 1 has 57% shorter endurance lifetime than SLC NAND flash in Example 2. This is because acceptable W/E cycle of TLC NAND flash is only 3% of that of SLC NAND flash. Thanks to large capacity, 9%, of SCM, TLC NAND flash in Example 3 achieves 94% longer endurance

lifetime than SLC NAND flash in Example 2. For hm_0, W/E cycle gap between top and bottom NAND flash is smaller than that of prxy_0. Due to less hot data in hm_0 than prxy_0, top NAND flash such as SLC and TLC NAND flash for hm_0 has longer endurance lifetime than that for prxy_0.

C. Performance and SSD endurance lifetime comparison with limited SSD cost

Because SSD requires both high performance and low cost, this paper limits the total SSD cost just like Case 1 and Case 2 as shown in Table 3. Furthermore, Fig. 11 shows the highest IOPS/cost in each configuration. Table 4 summarizes IOPS/cost of SCM/TLC/QLC NAND flash. For hm_0 and src2_1, SCM/TLC NAND flash has the highest IOPS/cost in both cases. Moreover, for prxy_0 and prxy_1, SCM/TLC/QLC NAND flash achieves the highest IOPS/cost in Case 1. Fig. 12 shows the largest SSD endurance lifetime. Note that "SSD endurance lifetime" is NOT equal to "memory endurance lifetime" discussed previously. Table 5 summarizes SSD endurance lifetime of SLC/QLC NAND flash for prxy_0 and hm_0. SSD endurance lifetime is defined as the shortest memory endurance lifetime in each configuration. For prxy_0, SLC/QLC has the longest SSD endurance lifetime in Case 1. For all workloads, SCM/TLC/QLC NAND flash has shorter SSD endurance lifetime than SCM/TLC NAND flash in both cases. This is because TLC NAND flash capacity of SCM/TLC/QLC NAND flash is less than half of that of SCM/TLC NAND flash and thus shortens SSD endurance lifetime.

V. CONCLUSION

This paper analyses SSD performance, energy consumption and SSD endurance lifetime of heterogeneous SSD in four configurations: 1) SLC/QLC NAND flash, 2) SCM/QLC NAND flash, 3) SCM/TLC/QLC NAND flash and 4) SCM/TLC NAND flash. Table 6 summarizes recommended configuration. In case of cold workloads like hm_0 and src2_1 or high acceptable cost (Case 2), SCM/TLC NAND flash is recommended. For hot workloads like prxy_0 and prxy_1 with low total SSD cost (Case 1), however, SLC/QLC NAND flash or SCM/TLC/QLC NAND flash is recommended. SLC/QLC NAND flash has better endurance lifetime. On the other hand, SCM/TLC/QLC achieves higher performance and lower energy consumption.

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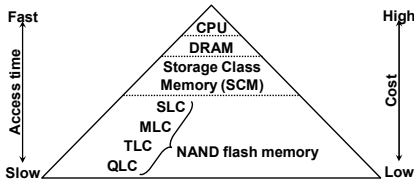


Fig. 1. Memory hierarchy.

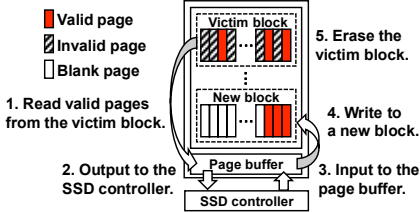


Fig. 3. Garbage collection (GC) operation in NAND flash memory.

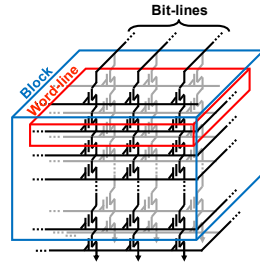


Fig. 2. Memory cell array of 3D NAND flash [5].

Table 1. Memory characteristics of SCM [8, 15], SLC [16], TLC and QLC [2, 17] NAND flash.

	SCM	SLC NAND flash	TLC NAND flash	QLC NAND flash
Write latency	100 ns/sector	160 us/page	2190 us/word-line (= 3 pages)	12410 us/word-line (= 4 pages)
Read latency	100 ns/sector	30 us/page	Average 66 us/page	Average 140 us/page
Erase latency	N/A	3 ms/block	4.8 ms/block	3.5 ms/block
Page size	N/A	16 KB	16 KB	16 KB
Block size	N/A	4.2 MB	12.6 MB	16.8 MB
Write unit	sector	page	word-line	block
V_{DD} (core / I/O)	1.8 V / 1.2 V	3.3 V / 1.8 V	3.3 V / 1.8 V	3.3 V / 1.8 V
Acceptable W/E cycles	10^{12}	10^5	3×10^3	10^3
Bit cost	20	3	1	3/4

Table 2. Characteristics of four types of workload.

Workload	User data size (GB)	Total write data size (GB)	Total read data size (GB)	Characteristic
prxy_0	0.98	54.05	3.05	write-hot
hm_0	2.54	21.67	9.96	write-cold
prxy_1	13.94	726.1	1294	read-hot
src2_1	19.27	19.38	37.02	read-cold

Write-intensive: total write access size > Total read access size
Read-intensive: total write access size < Total read access size
Hot: total write access size/user data size > 40
or total read access size/user data size > 40
Cold: total write access size/user data size < 40
or total read access size/user data size < 40

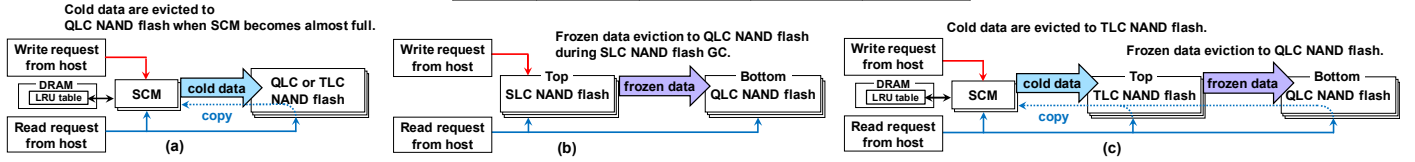


Fig. 4. (a) NV-WB cache algorithm [11] and (b) RR-FDCA [12]. NV-WB is applied to SCM and NAND flash. RR-FDCA is applied to two types of NAND flash. Algorithm combining NV-WB cache algorithm and RR-FDCA (c) is applied to SCM/TLC/QLC NAND flash [13].

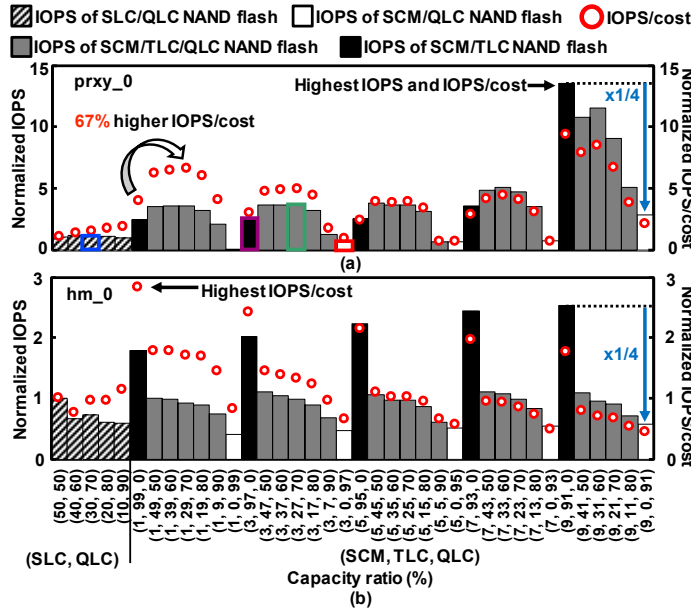


Fig. 5. IOPS and IOPS/cost of SLC/QLC, SCM/QLC, SCM/TLC/QLC and SCM/TLC NAND flash for (a) prxy_0 and (b) hm_0.

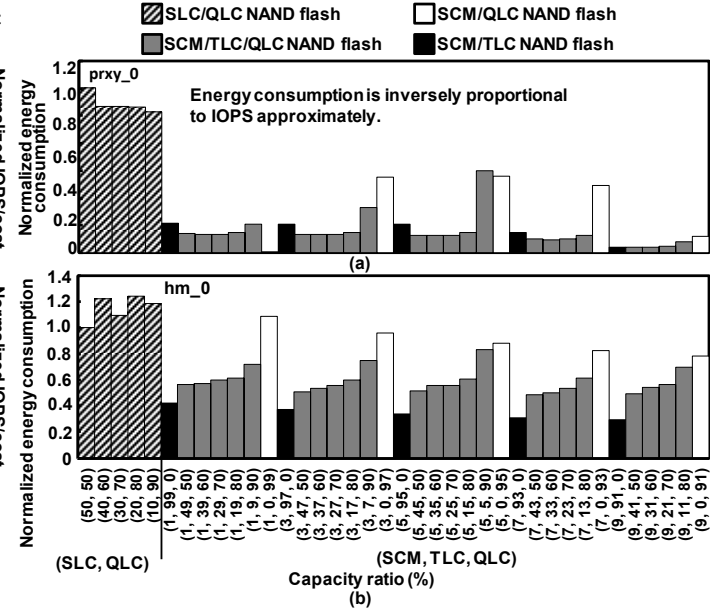


Fig. 6. Energy consumption of SLC/QLC, SCM/QLC, SCM/TLC/QLC and SCM/TLC NAND flash for (a) prxy_0 and (b) hm_0.

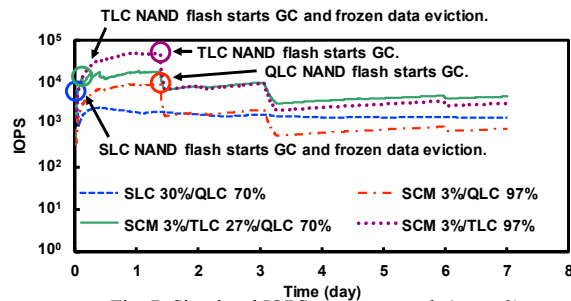


Fig. 7. Simulated IOPS over one-week (prxy_0).

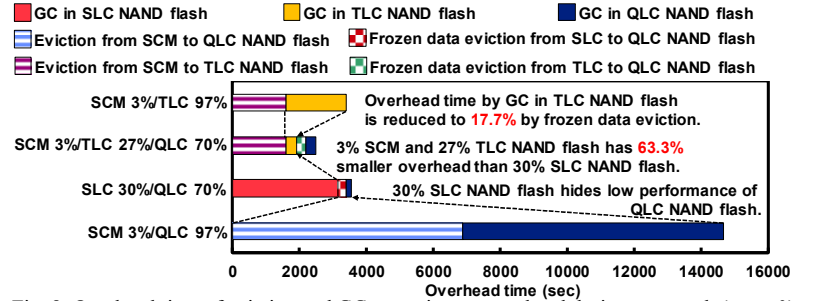


Fig. 8. Overhead time of eviction and GC operation accumulated during one-week (prxy_0).

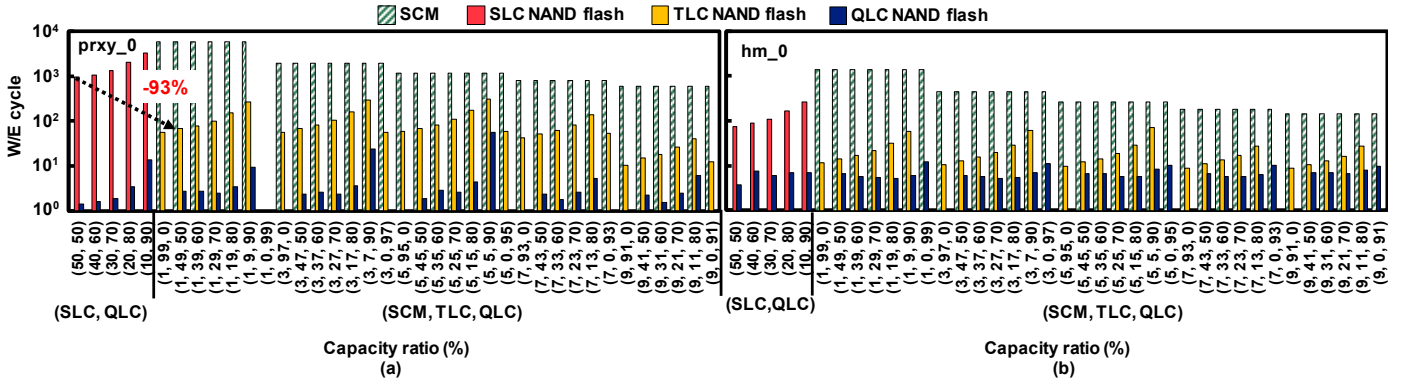


Fig. 9. W/E cycles of SCM, SLC, TLC and QLC NAND flash for (a) prxy_0 and (b) hm_0.

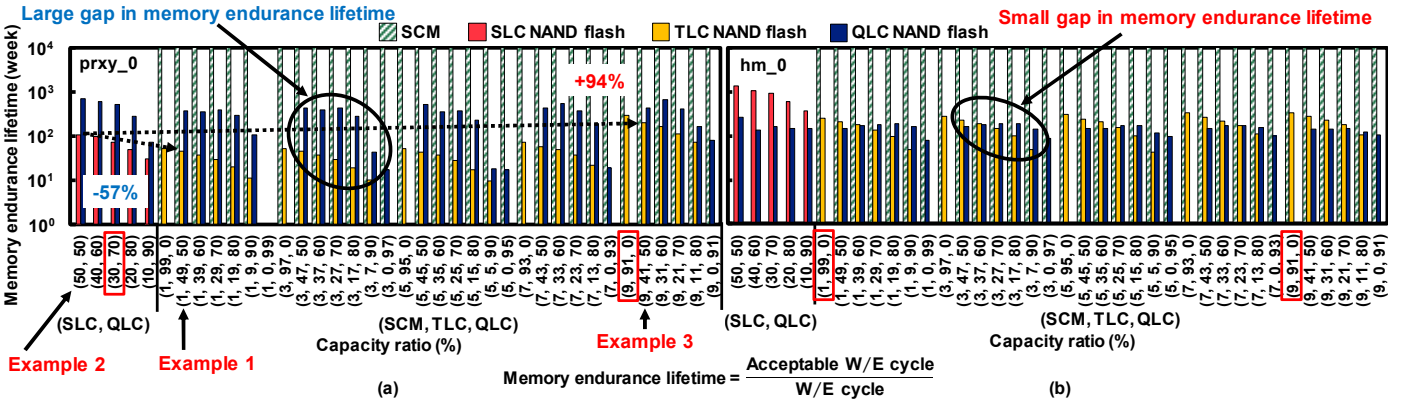


Fig. 10. Memory endurance lifetime of SCM, SLC, TLC and QLC NAND flash for (a) prxy_0 and (b) hm_0.

Table 3. Assumption of acceptable cost.

	Acceptable total SSD cost
Case 1	1.5x TLC NAND flash only SSD
Case 2	3.0x TLC NAND flash only SSD

Table 4. Highest IOPS/cost of SCM/TLC/QLC NAND flash in each case and workload.

Case	Workload	Normalized by SLC/QLC	Normalized by SCM/QLC	Normalized by SCM/TLC
Case 1	prxy_0 (write-hot)	x3.59	x7.39	x1.67
	hm_0 (write-cold)	x1.56	x2.18	x0.630
	prxy_1 (read-hot)	x1.27	x5.87	x1.65
	src2_1 (read-cold)	x1.34	x1.38	x0.469
Case 2	prxy_0 (write-hot)	x4.63	x4.01	x0.902
	hm_0 (write-cold)	x1.56	x2.18	x0.630
	prxy_1 (read-hot)	x3.64	x1.52	x0.479
	src2_1 (read-cold)	x1.26	x1.38	x0.469

Table 5. Longest SSD endurance lifetime of SLC/QLC NAND flash in each case.

Case	Workload	Normalized by SCM/QLC	Normalized by SCM/TLC/QLC	Normalized by SCM/TLC
Case 1	prxy_0 (write-hot)	x4.18	x1.63	x1.38
	hm_0 (write-cold)	x1.87	x0.939	x0.656
Case 2	prxy_0 (write-hot)	x1.32	x0.515	x0.357
	hm_0 (write-cold)	x2.55	x1.50	x0.775

Table 6. Recommended memory configuration from the perspective of performance, energy consumption and SSD endurance lifetime.

Workload	Hot (prxy0, prxy_1)		Cold (hm_0, src2_1)
Acceptable total SSD cost	Case 1	Case 2	Case 1, 2
Performance and energy consumption	SCM/TLC/QLC	SCM/TLC	SCM/TLC
SSD endurance lifetime	SLC/QLC	SCM/TLC	SCM/TLC

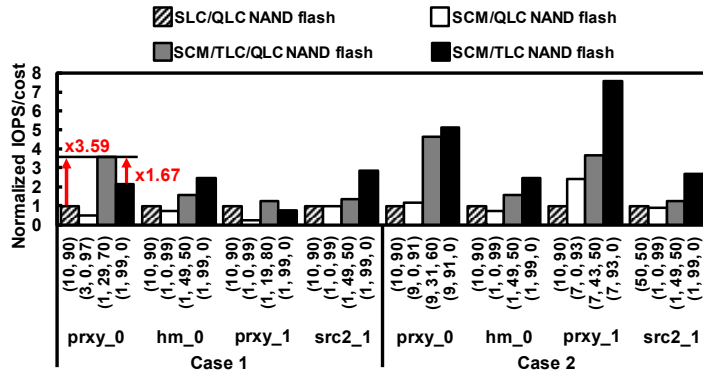


Fig. 11. Highest (a) IOPS and (b) IOPS/cost in each case, workload and memory configurations.

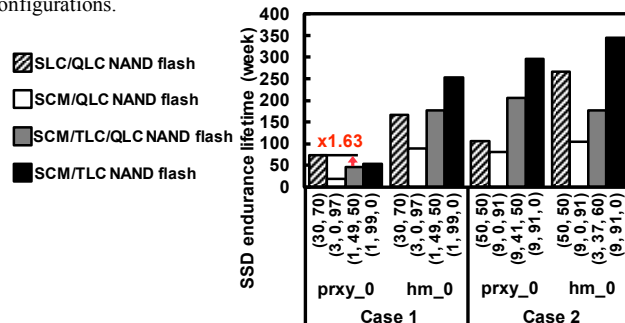


Fig. 12. Longest SSD endurance lifetime in each case and memory configurations for prxy_0 and hm_0.